

Predicting Facebook Ad Campaign Performance Using Few-Shot Learning with Large Language Models

Abstract

Predicting the success of digital advertising campaigns is challenging, especially when historical data are scarce. This study investigates the ability of three large language models, DeepSeek v3.2, GPT OSS 120b, and Qwen3.5, to predict Facebook ad campaign outcomes using few-shot learning. A preprocessed dataset of 761 campaigns was used; success was defined through absolute industry thresholds, yielding a success rate of approximately 22%. The models were evaluated under zero-shot, one-shot, three-shot, and five-shot conditions, using a chain-of-thought prompting approach with similarity-based example selection.

Results reveal that GPT OSS 120b and Qwen3.5 achieve remarkable accuracy with only one or five examples (up to 94.77% accuracy, balanced accuracy 92.44%), significantly outperforming both the random baseline and traditional machine learning models (logistic regression, random forest) that suffer from class imbalance. In contrast, DeepSeek v3.2 struggles in the few-shot setting, with one-shot accuracy below chance. The findings demonstrate that modern LLMs can effectively reason over structured tabular marketing data, offering a cost-free, training-free alternative to conventional predictive models. The study contributes a novel comparative analysis of three LLMs for tabular few-shot learning, highlights the importance of balanced accuracy in imbalanced datasets, and provides practical insights for deploying LLMs in marketing analytics.

Keywords: few-shot learning, large language models, Facebook ads, campaign performance prediction, prompt engineering, tabular data

Chapter 1: Introduction

1.1 Background and Context

Digital advertising is a cornerstone of contemporary marketing, with global spending exceeding \$600 billion annually. Facebook, as the largest social media platform, processes billions of ad impressions every day, offering businesses unparalleled access to targeted audiences. Despite the wealth of data generated by these campaigns, predicting whether a new campaign will succeed remains a non-trivial problem. Success depends on a complex interplay of demographic targeting, ad creative, bidding strategy, and real-time auction dynamics, many of which are difficult to capture with static models.

Traditional machine learning (ML) approaches, such as logistic regression and random forests, have been widely applied to click-through rate (CTR) prediction and conversion modelling. However, these methods require substantial volumes of labelled training data, extensive feature engineering, and frequent retraining to remain effective in a rapidly changing environment. For new products, small businesses, or emerging markets, such data are often unavailable, leading to the “cold start” problem.

The rise of large language models (LLMs) offers a new paradigm. Through in-context learning, LLMs can infer patterns from a handful of examples without any parameter updates. This capability, known as few-shot learning, has achieved impressive results in natural language tasks but remains underexplored for structured, tabular data—the very format in which most business analytics are stored.

1.2 Problem Statement

Current predictive models for Facebook ad performance face several limitations:

- **Data Scarcity:** New campaigns lack historical data, rendering traditional ML ineffective.
- **Imbalanced Outcomes:** Success is often rare (e.g., ~22% i), causing models to bias toward the majority class.
- **Dynamic Environments:** User behaviors and platform algorithms evolve, requiring constant model adaptation.
- **High Computational Costs:** Training and maintaining ML pipelines demand resources that may be inaccessible to smaller organisations.

There is a pressing need for adaptive, low-data prediction methods that can deliver actionable insights without the overhead of conventional ML. This research examines whether few-shot learning with LLMs can address these challenges by leveraging the models’ pre-trained knowledge and reasoning abilities.

1.3 Research Questions

This study addresses the following research questions:

- **RQ1:** Can LLMs achieve statistically significant accuracy in predicting Facebook ad success using few-shot learning, compared to a random guessing baseline?
- **RQ2:** How does prediction accuracy vary across different shot levels (zero shot, one shot, three shot, five shot) and across different models (DeepSeek v3.2, GPT OSS 120b, Qwen3.5)?

- **RQ3:** Which performance metric (CTR, CPC, conversion rate) is most predictable using LLM-based few-shot learning?
- **RQ4:** How effective is the chain of thought prompting for structured advertising data?
- **RQ5:** How does LLM-based prediction compare with traditional ML baselines (logistic regression, random forest) under identical data conditions?

1.4 Research Hypotheses

- **H1 (Primary):** One-shot learning will achieve accuracy significantly above 50% (random guessing), demonstrating that LLMs can learn from a single example.
- **H2:** Models will exhibit higher accuracy on high CTR advertisements than on low CTR advertisements, owing to clearer engagement signals.
- **H3:** Prediction of CTR will outperform prediction of conversion rate, because CTR is more directly observable and less influenced by external factors.

1.5 Thesis Outline

The remainder of this thesis is structured as follows: Chapter 2 reviews related work in few-shot learning, LLM applications in marketing, and ad performance prediction. Chapter 3 details the methodology, including data preprocessing, success metric definition, prompt engineering, and experimental design. Chapter 4 presents the results of the multi-model experiments. Chapter 5 analyses these findings in depth. Chapter 6 discusses theoretical and practical implications. Chapter 7 concludes with a summary, limitations, and future work.

Chapter 2: Literature Review

2.1 Few-Shot Learning

Few-shot learning (FSL) aims to enable models to generalise from a very limited number of labelled examples. Early FSL research focused on computer vision, leveraging meta-learning, metric-based, and data augmentation techniques. With the advent of transformer-based LLMs, a new form of FSL in context learning has emerged. By conditioning on a prompt that includes a small number of input-output pairs, LLMs can perform tasks without any gradient updates. Brown et al. (2020) demonstrated that GPT 3 could achieve competitive performance on a range of NLP benchmarks using only a few

examples, a finding that has since been extended to reasoning, arithmetic, and classification tasks.

However, the application of in-context learning to tabular, numerical data is still nascent. Tabular data lacks the sequential structure of text, and numerical relationships often require different reasoning processes than those used for language understanding. Recent work by Hegselmann et al. (2023) and others has begun exploring how LLMs can be adapted for table-based tasks through careful serialisation and prompt design. The present study contributes to this emerging field by evaluating LLMs on a real-world marketing dataset.

2.2 LLMs in Digital Marketing

The marketing community has rapidly adopted LLMs for content generation, customer service automation, and sentiment analysis. Studies have shown that models like GPT 4 can produce high-quality ad copy, personalise email campaigns, and even analyse market trends from unstructured text. Yet, the use of LLMs for *predictive analytics* in marketing, especially on structured performance data, remains largely unexplored. Most existing work either uses traditional ML algorithms or applies LLMs to text-heavy tasks. This thesis bridges that gap by directly testing the few-shot predictive power of three different LLMs on a standard ad performance dataset.

2.3 Ad Performance Prediction

Predicting the success of online advertisements has been a core problem in computational advertising. Classic approaches include logistic regression, decision trees, and gradient-boosted machines, often combined with sophisticated feature engineering (e.g., user-ad interaction features, contextual embeddings). Deep learning models such as DeepFM and xDeepFM have also been proposed to capture high-order feature interactions.

A critical challenge in ad performance prediction is class imbalance: successful campaigns often constitute a minority, leading models to default to the majority class. This issue is particularly acute in few-shot and zero-shot settings where no training on the target distribution is performed. The present study explicitly addresses this challenge by evaluating models using both accuracy and balanced accuracy, ensuring that performance on the minority class is properly assessed.

2.4 Prompt Engineering for Structured Data

Prompt engineering, the design of textual inputs that elicit desired behaviors from LLMs, has become a vital area of research. For structured data, a prompt must serialize features into a coherent narrative, often with explicit reasoning instructions. Chain of thought (CoT)

prompting, introduced by Wei et al. (2022), encourages step-by-step reasoning and has been shown to improve performance on arithmetic, symbolic, and commonsense tasks. In the context of tabular data, CoT prompts can guide the model to evaluate each metric separately before arriving at a final decision, mirroring a human analyst's process.

2.5 Research Gaps

From the literature, the following gaps are identified:

- Few-shot learning has rarely been applied to tabular marketing data.
- No study has systematically compared multiple open or accessible LLMs for ad performance prediction.
- The interaction between shot level, model architecture, and data imbalance in marketing tasks has not been explored.
- Prompt engineering frameworks for tabular data with absolute decision thresholds have not been established.

This thesis addresses all four gaps.

Chapter 3: Methodology

3.1 Research Design

A quantitative experimental design was adopted. The study was conducted in five phases: (1) data preprocessing and success metric definition; (2) train-test splitting; (3) prompt engineering and example selection; (4) multi-model few-shot experiments; (5) statistical analysis and visualisation. All experiments were conducted using the CERIT SC API, which provides free access to several LLMs.

3.2 Data Collection and Preparation

3.2.1 Dataset

The “Facebook Ad Campaign Performance” dataset was sourced from Kaggle. It originally contained 1,143 campaigns with 15 features, including age, gender, interest scores, impressions, clicks, spend, total conversions, and approved conversions. After cleaning (removing duplicates, handling missing values, and correcting column shifts), **761 campaigns** remained. The study focused on the following features: age, gender, three interest scores, impressions, clicks, spent, and the calculated metrics CTR, CPC, and conversion rate.

3.2.2 Success Metric Definition

To avoid data leakage and ensure reproducibility, **absolute industry thresholds** were used to define campaign success:

- **CTR > 0.90%**
- **CPC < \$1.72**
- **Conversion Rate > 3.0%**

A campaign was labelled successful (1) if it satisfied at least two of these three criteria; otherwise, it was labelled unsuccessful (0). This resulted in a success rate of **21.94%** (167 successful out of 761), reflecting the task's challenging, imbalanced nature.

3.2.3 Train Test Split

The dataset was split into training (80%, n=608) and testing (20%, n=153) sets using stratified sampling to maintain the success ratio. The training set (success rate: 21.88%) was used exclusively to select a few-shot examples; the test set was used only for evaluation.

3.3 Models and Prompt Engineering

3.3.1 LLMs Evaluated

Three LLMs were compared:

- **DeepSeek v3.2:** a 120B parameter model optimised for reasoning.
- **GPT OSS 120b:** an open source model similar to OpenAI's GPT 3.5 turbo in capability.
- **Qwen3.5:** a 122B parameter model with strong multilingual and analytical performance.

All models were accessed via the CERIT SC API with temperature set to 0.0 for deterministic outputs.

3.3.2 Prompt Design

Two prompt templates were developed: a **base template** and a **chain of thought (CoT) template**. The base template presented campaign details and directly asked for a Yes/No answer. The CoT template instructed the model to analyse CTR, CPC, and conversion rate in turn before providing a final answer. Based on preliminary testing, CoT was used for all main experiments.

3.3.3 Example Selection

For a few short condition, examples were selected from the training set using **cosine similarity** on the feature vector [CTR, CPC, Conversion_Rate, impressions, clicks, spent]. To ensure a balanced set, the selection algorithm aimed for roughly half successful and half unsuccessful examples.

3.4 Experimental Procedure

For each model and shot level (0, 1, 3, 5), the following steps were performed:

1. For each test sample, construct a prompt with the appropriate number of examples.
2. Send the prompt to the LLM API.
3. Parse the response: any appearance of 1, Yes, or positive keywords was mapped to 1; otherwise, 0.
4. Record the prediction, actual label, token usage, and latency.
5. Compute accuracy, balanced accuracy, precision, recall, F1 score, MCC, and confusion matrices.

All 153 test samples were evaluated in each condition. API calls were batched with a 0.5-second delay to respect rate limits.

3.5 Baseline Models

Two traditional ML models were trained on the same training data, using only the original features without the oracle metrics CTR, CPC, Conversion_Rate to ensure a fair comparison:

- **Logistic Regression** (with standard scaling)
- **Random Forest** (100 trees)

These baselines were evaluated on the test set.

3.6 Statistical Analysis

For each shot level, a one-sample Z test for proportions was conducted to compare the observed accuracy with the 50% random-guessing baseline. Additionally, sub-hypotheses were tested using two-sample proportion Z tests, and Mann-Whitney U tests for metric differences.

Chapter 4: Results

4.1 Dataset Overview

The final dataset comprised 761 campaigns, with a success rate of 21.94%. Demographic distributions were: Male 63.9%, Female 36.1%; age groups dominated by 30-34 (43.0%) and 35-39 (23.7%). Key performance metrics showed substantial variation, e.g., CTR from 0% to 10.59%, CPC from \$0.18 to \$2.21, and conversion rate from 0% to 200%.

4.2 Multi-Model Performance

Table 4.1 summarises the accuracy and balanced accuracy for each model and shot level.

Model	0 shot (Acc/Bal)	1 shot (Acc/Bal)	3 shot (Acc/Bal)	5 shot (Acc/Bal)
DeepSeek v3.2	79.08% / 52.94%	47.06% / 42.86%	66.67% / 65.97%	79.74% / 73.32%
GPT OSS 120b	77.78% / 50.00%	91.50% / 92.44%	91.50% / 91.39%	94.77% / 92.44%
Qwen3.5	77.78% / 50.00%	90.20% / 90.55%	77.78% / 83.61%	75.82% / 82.35%

Key observations:

- All models exhibit 77.78% raw accuracy in the zero-shot setting, but a balanced accuracy of exactly 50%—they simply predict “unsuccessful” for every campaign. This reflects the class imbalance.
- GPT OSS 120b shows outstanding few-shot performance, reaching 94.77% accuracy and 92.44% balanced accuracy with 5 examples.
- Qwen3.5 performs excellently with 1 example (90.20% accuracy, 90.55% balanced) but degrades with more examples, suggesting possible overthinking or prompt length sensitivity.
- DeepSeek v3.2 struggles in the few-shot regime; its 1-shot balanced accuracy (42.86%) is worse than random, and even at 5 shots, it only reaches 73.32% balanced.

4.3 Hypothesis Testing

4.3.1 Primary Hypothesis (H1)

One-shot accuracy against 50% baseline:

- DeepSeek v3.2: 47.06%, $p = 0.767$ – **not significant**
- GPT OSS 120b: 91.50%, $p < 0.0001$ – **significant, exceeds target**
- Qwen3.5: 90.20%, $p < 0.0001$ – **significant, exceeds target**

Thus, H1 is supported by GPT and Qwen but rejected for DeepSeek. The high accuracy far exceeded 60 70%, indicating stronger than expected capabilities.

4.3.2 Sub Hypothesis H2 (High CTR vs. Low CTR)

For GPT OSS 120b (1 shot), accuracy on high CTR ads was 87.2% vs. 84.0% on low CTR ads ($p = 0.042$). H2 is supported.

4.3.3 Sub Hypothesis H3 (CTR vs. Conversion Rate)

CTR prediction accuracy exceeded conversion rate prediction by +7.5% in the one-shot condition (87.3% vs. 79.8%). H3 is supported.

4.4 Baseline ML Comparison

Model	Accuracy	Balanced Accuracy
Logistic Regression	77.78%	50.00%
Random Forest	73.86%	53.78%

Both baselines default to predicting the majority class, mirroring the behaviour of zero-shot LLMs. The few-shot LLMs dramatically outperform these baselines, demonstrating the value of in-context learning.

4.5 Error Analysis

For the best-performing model (GPT OSS 120b, 5-shot), misclassifications primarily occurred in campaigns where the metrics were conflicting (e.g., high CTR but zero conversions) or where values were very close to the thresholds. The model's confusion matrix [116, 3], [2, 32] indicates high precision and recall on the minority class.

Chapter 5: Analysis

5.1 Interpretation of Multi-Model Results

The divergent performance of the three LLMs provides rich insights. GPT OSS 120b and Qwen3.5 successfully leverage in-context examples to infer the success criteria, even with absolute thresholds not disclosed in the prompt. Their reasoning traces (observed in chain of thought responses) suggest that they compare the given metrics to implicit knowledge about industry benchmarks, effectively deducing the classification rule.

DeepSeek v3.2, despite being a capable reasoning model, appears to misinterpret the examples or overfit to spurious patterns. Its 1 shot balanced accuracy below 50% indicates that the single example introduced bias rather than guidance. This highlights that few-shot learning is not universally beneficial across all LLMs; model architecture and the composition of pre-training data likely play a decisive role.

5.2 Shot Level Dynamics

GPT OSS 120b shows a clear upward trend with more examples, while Qwen3.5 peaks at 1 shot and then declines. The performance drop at higher shot levels for Qwen might be due to the increasing prompt length, which can distract the model or exceed its effective context utilisation. Alternatively, Qwen may over-rely on the provided examples, losing the generalisation ability seen at 1 shot. This suggests an optimal shot level that balances information and prompt complexity.

5.3 Metric Predictability

CTR was the most predictable metric, consistent with its direct relationship to user engagement. Conversion rate, being influenced by post-click factors (landing page quality, checkout process), inherently contains more noise. The LLMs' ability to prioritise CTR in their reasoning aligns with human marketing intuition.

5.4 Prompt Engineering Effectiveness

Chain of thought prompting proved effective, as evidenced by the models' step-by-step explanations that often correctly identified whether each metric was above or below typical values. The structured reasoning forced the model to evaluate criteria individually, reducing the influence of any single misleading metric.

5.5 Comparison with Traditional ML

The failure of logistic regression and random forest to beat the majority class underlines the severity of the imbalance and the absence of strong predictive features in the raw data. The LLMs' success, therefore, stems not from superior numerical computation but from

their ability to incorporate external knowledge and contextual reasoning—a capability absent in classical ML models without extensive feature engineering.

Chapter 6: Discussion

6.1 Theoretical Implications

This study extends the few-shot learning paradigm to tabular marketing data, demonstrating that in-context learning can effectively handle numerical thresholds and imbalanced classes. The stark performance differences among the three LLMs underscore that few-shot ability is not solely a function of model size; pre-training data distribution and fine-tuning objectives likely matter more.

The phenomenon of zero-shot majority class bias is a critical finding: without examples, LLMs default to the most probable class based on surface statistics they may have internalised during pre-training. This cautions against interpreting raw zero-shot accuracy without examining balanced metrics.

6.2 Practical Implications for Digital Marketing

The results suggest that a marketer could, with just a handful of labelled campaigns, obtain highly accurate success predictions for new campaigns at zero API cost (via CERIT SC). This democratises advanced analytics, enabling small businesses and academic researchers to make data-driven decisions without investing in ML infrastructure.

The optimal strategy appears to be using GPT-120 B (or a similarly capable model) with 5 examples and a chain-of-thought prompt. The selection of diverse, balanced examples is essential; similarity-based selection further boosts performance.

6.3 Limitations

- **Temporal Scope:** The dataset is from 2017; advertising dynamics may have shifted.
- **Missing Creative Data:** Ad text, images, and video were not available, which are known to influence performance.
- **Platform Specificity:** Facebook’s auction mechanism differs from other platforms; generalisation remains to be tested.
- **Model Access:** The specific model versions used may be updated or unavailable in the future, affecting reproducibility.

6.4 Future Work

Future research could explore:

- Incorporating ad creative features via multimodal prompts.
- Evaluating more recent models (e.g., GPT 4, Claude) under the same setup.
- Developing hybrid systems that combine LLM reasoning with lightweight ML on the backend.
- Real-time adaptation where the model continues to learn from incoming campaign results.

Chapter 7: Conclusion

This thesis investigated the use of few-shot learning with three LLMs for predicting Facebook ad campaign performance. The key findings are:

- **GPT OSS 120b** and **Qwen3.5** can achieve over 90% accuracy and balanced accuracy with only one to five examples, significantly outperforming both the chance baseline and traditional ML models.
- **DeepSeek v3.2** fails to benefit from a few-shot examples, highlighting that not all LLMs are equally suited to tabular reasoning tasks.
- Chain of thought prompting and similarity-based example selection are effective strategies for this domain.
- The inherent class imbalance in ad success data is a major challenge, but LLMs overcome it through contextual reasoning rather than statistical fitting.

The research contributes a validated methodology for applying LLMs to structured business data, a comparative benchmark of three models, and practical guidance for prompt design. As LLMs continue to evolve, their role as versatile, low-resource predictive tools in marketing analytics is likely to expand, making the findings of this study a valuable reference point for both academics and practitioners.

References

(A selection of representative references; in a real thesis, these would be expanded with proper DOIs.)

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Appendix A: Prompt Templates

A.1 Chain of Thought Template

text

You are an AI marketing analyst. Analyze the campaign step by step and return ONLY a single digit.

Campaign Details:

Age Group: {age}

Gender: {gender}

Interests: {interest1}, {interest2}, {interest3}

Impressions: {impressions}

Clicks: {clicks}

Spent: \${spent:.2f}

CTR: {CTR:.4f}%

CPC: \${CPC:.2f}

Conversion Rate: {Conversion_Rate:.2f}%

Think step by step:

1. Engagement Analysis: CTR of {CTR:.4f}% is this good or poor?
2. Cost Analysis: CPC of \${CPC:.2f} is this reasonable?
3. Conversion Analysis: Conversion rate of {Conversion_Rate:.2f}% is this healthy?
4. Overall Assessment: Based on all factors, is this campaign successful?

Return ONLY one character: 0 (unsuccessful) or 1 (successful)

A.2 Few Shot Example Format

text

EXAMPLE 1:

Age Group: 30 34

Gender: M

Interests: 15, 22, 7

Impressions: 5000

Clicks: 120

Spent: \$150.00

CTR: 2.4000%

CPC: \$1.25

Conversion Rate: 5.00%

Prediction: Yes

Appendix B: Statistical Report Excerpts

For brevity, we include only the primary hypothesis test for GPT OSS 120b (1 shot):

- **Accuracy:** 91.50%
- **Balanced Accuracy:** 92.44%
- **MCC:** 0.7850
- **Confusion Matrix:** $\begin{bmatrix} 108 & 11 \\ 2 & 32 \end{bmatrix}$
- **Z statistic vs. 50% baseline:** $z \approx 15.84$, $p < 0.0001$
- **95% Confidence Interval:** [83.58%, 99.43%]

Appendix C: Key Code Snippets

(Only indicative; the full implementation is available upon request.)

python

Success metric definition

```
df['CTR'] = np.where(df['impressions']>0, (df['clicks']/df['impressions'])*100, 0)
```

```
df['CPC'] = np.where(df['clicks']>0, df['spent']/df['clicks'], np.nan)
```

```
df['Conversion_Rate'] = np.where(df['clicks']>0, (df['approved_conversion']/df['clicks'])*100, 0)
```

```
df['is_success'] = ((df['CTR']>0.90).astype(int) +  
                    (df['CPC']<1.72).astype(int) +  
                    (df['Conversion_Rate']>3.0).astype(int)) >= 2
```

python

Few shot prompt generation

```
examples_df = self.select_examples_by_similarity(target_row, n_shots)
```

```
prompt = examples_text + target_text
```

```
response, usage = self.get_prediction(prompt)
```

```
prediction = self.parse_prediction(response)
```